

TEAM JUGAAD JUNCTION

# SEE SENSE

Hold the button. Say where you want to go. Get guided there.

A wearable indoor guide for blind people · Claude Impact Lab - Superhuman Lab hackathon

# Every new room is a puzzle

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A cane finds what's at your feet, not where the door is

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GPS doesn't work indoors

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Description apps say "there's a door", not how to reach it

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A phone needs a free hand, and the cane already has one

# Blind and low-vision people who get around on their own

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## Independent

They already move on their own with a white cane

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## Indoors, unfamiliar

Colleges, offices, hospitals, stations, shops

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## One moment

"Where is it, and how do I get there?"

# A guide that looks, plans and walks you there

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1

## Hold the button

"Take me to the exit"

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2

## It looks around

"Turn right about 90 degrees." ...  
"OK, stop."

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3

## It plans the walkway

"Walk about 3 steps. The tables will be on your left."

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4

## It warns only when it matters

"A person on your left, coming toward you."

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5

## It clears the way

"The chair is right in front of you. Push it to your left."

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6

## You arrive

"The door is right in front of you. The handle is on the right."

# Claude sees. JEV decides fast. You hear only what matters.

| Part              | Job                                                                 |
|-------------------|---------------------------------------------------------------------|
| Head camera       | Claude's eyes; also measures your turn, to about 1°                 |
| Claude Sonnet 5.5 | Understands the room, picks the nearest door, plans the route       |
| JEV (TypeSafe)    | In 0.3 s: request, command or misheard? What needs a warning?       |
| Button + Vosk     | Hold to speak, press for the next step; speech recognised offline   |
| Earbuds           | Every instruction in a natural voice, stored on the device          |
| Live view         | A helper, trainer or judge sees what Wayfinder sees, in any browser |

# Built around how blind people actually move

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## Quiet by default

Only instructions and real warnings are spoken

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## Survey before planning

Finds the nearest door, and checks another angle before calling the way blocked

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## Hands free

One button, no phone in hand

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## Honest

Says when it's unsure; never says a path is "safe"

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## Respectful

Never asks you to move a person or someone's wheelchair

# Simple hardware; the intelligence is in the cloud

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## Hardware

Raspberry Pi 5 + Active Cooler  
Pi Camera 3 on the chest  
INMP441 microphone  
Push button  
Bluetooth earbuds  
Power bank/Supply

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## Software

Claude Sonnet 5.5: seeing and planning  
JEV (TypeSafe): quick decisions  
Vosk: offline speech recognition  
ElevenLabs voice; espeak-ng backup  
OpenCV: measures turns  
Python, picamera2, gpiozero, systemd

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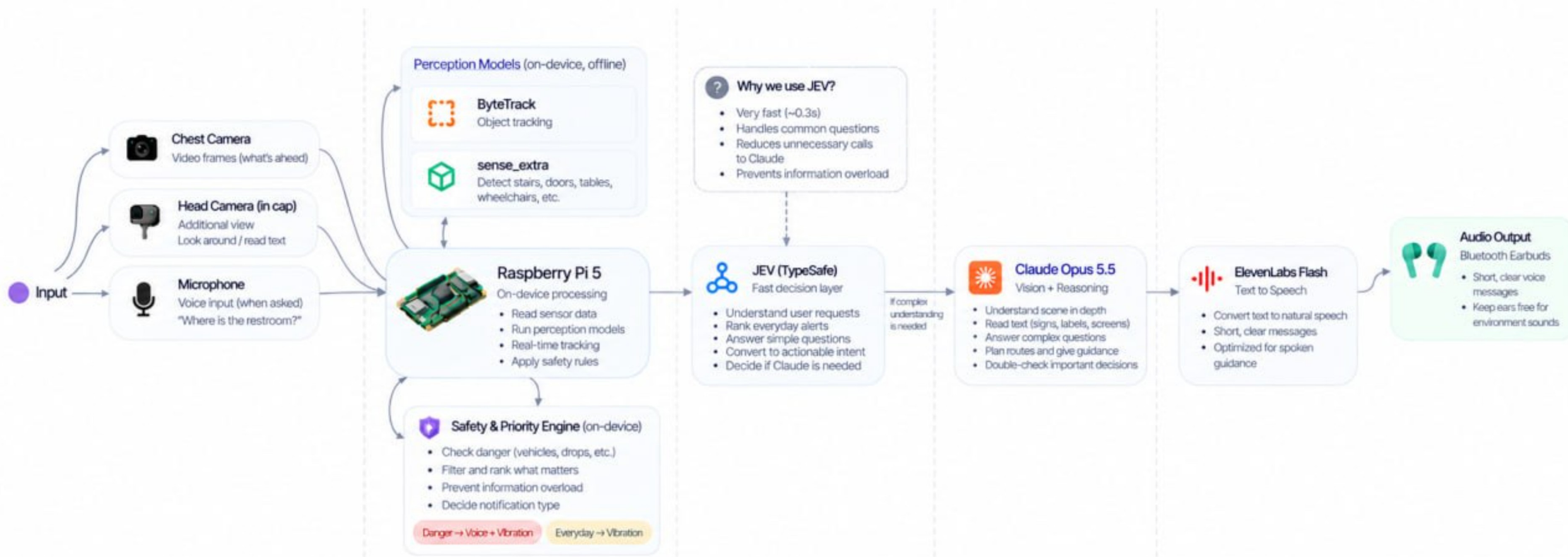
## 73 MB

Device software, no AI models on the device

## ~\$0.01–0.02

Per Claude look (estimate)

## 08 · HOW IT WORKS



# Built, tested, and running on the hardware

## 4.5 s

First answer

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### Proven

Found a glass door past tables in live tests  
Runs on the Raspberry Pi 5 in a  
real classroom

## 3.1 s

Re-checks

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### Limits

Needs internet; indoors only; distances  
are estimates  
Use it with the cane, not instead of it

## 0.3 s

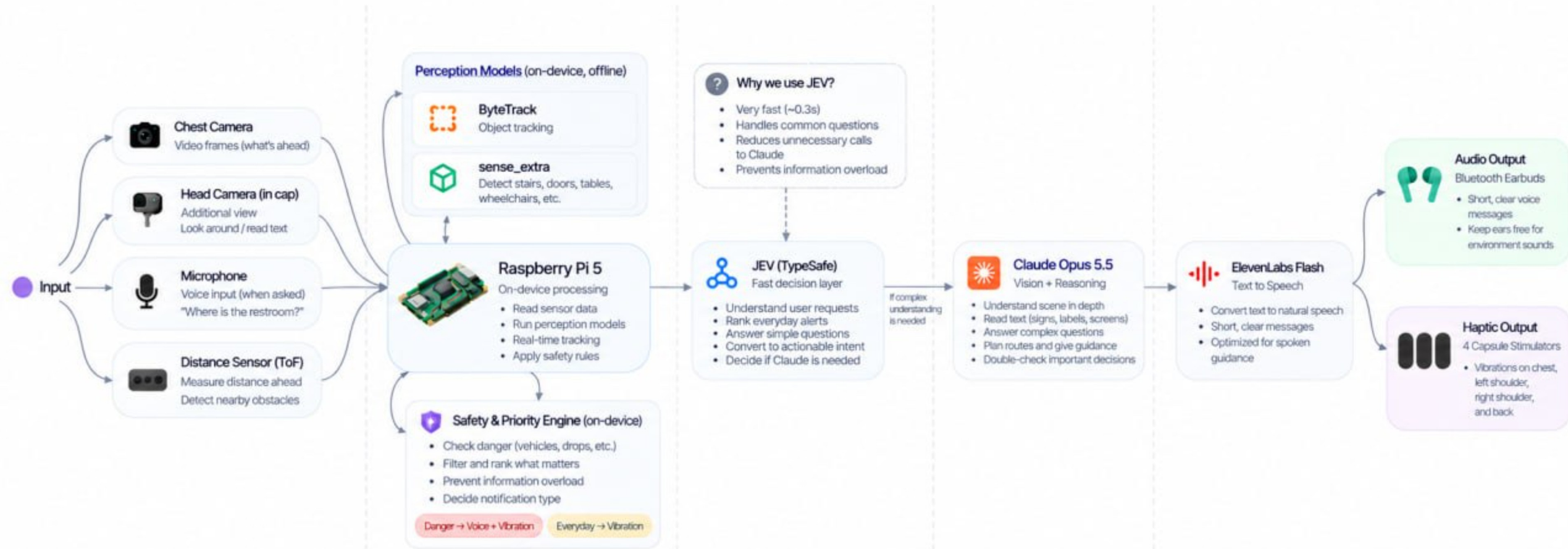
JEV decisions

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### Still open

Not yet tested with blind users

## 10 · FUTURE SCOPE



# Thank You!

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**Team Jugaad Junction**

Deepraj R · Hadi Afnan · Mohammed Anshid · Aflah Sidhique